SQL Database on Cloud:
AWS Relational Database Service (RDS)

1. Create MySQL Database on AWS RDS

Go to: https://console.aws.amazon.com

Then navigate to: **AWS Console → Services → RDS**

aws | Search [Alt+S] | Asia Pacific (Mumbai) | admin

S3

Aurora and RDS > Dashboard

Aurora and RDS

- Dashboard
- Databases
- Query editor
- Performance insights
- Snapshots
- Exports in Amazon S3
- Automated backups
- Reserved instances
- Proxies

- Subnet groups
- Parameter groups
- Option groups
- Custom engine versions
- Zero-ETL integrations

- Events
- Event subscriptions

- Recommendations 0
- Certificate update

Resources

Refresh

You are using the following Amazon RDS resources in the Asia Pacific (Mumbai) region (used/quota)

- DB Instances (0/20)
  - Allocated storage (0 TB/100 TB)
  - Instances and storage include Neptune and DocumentDB. Increase DB instances limit
- DB Clusters (0/40)
- Reserved instances (0/20)
- Snapshots (0)
  - Manual
    - DB Cluster (0/100)
    - DB Instance (0/100)
  - Automated
    - DB Cluster (0)
    - DB Instance (0)
- Recent events (0)
- Event subscriptions (0/20)
- Parameter groups (0)
  - Default (0)
  - Custom (0/40)
- Option groups (0)
  - Default (0)
  - Custom (0/20)
- Subnet groups (0/20)
- Supported platforms VPC
- Default network vpc-05d5014b44e8bfb33

Create a database

Amazon Relational Database Service (RDS) makes it easy to set up, operate, and scale a relational database in the cloud.

Create a database

You can use a backup from Amazon S3 to restore and create a new Aurora MySQL and MySQL database.

Restore from S3

Note: your DB instances will launch in the **Asia Pacific (Mumbai)** region

Explore Aurora & RDS

In this activity, you will learn how to create a database. To begin, choose **Start tutorial**.

Estimated duration
2-5 minutes

Start tutorial

RDS costs

Date range: Past 6 months

Region: Global

Total cost
$0.00

Regions
0

Cost ($)

Tutorials

Welcome to Aurora & RDS

The following tutorial will help you get started.

Create database

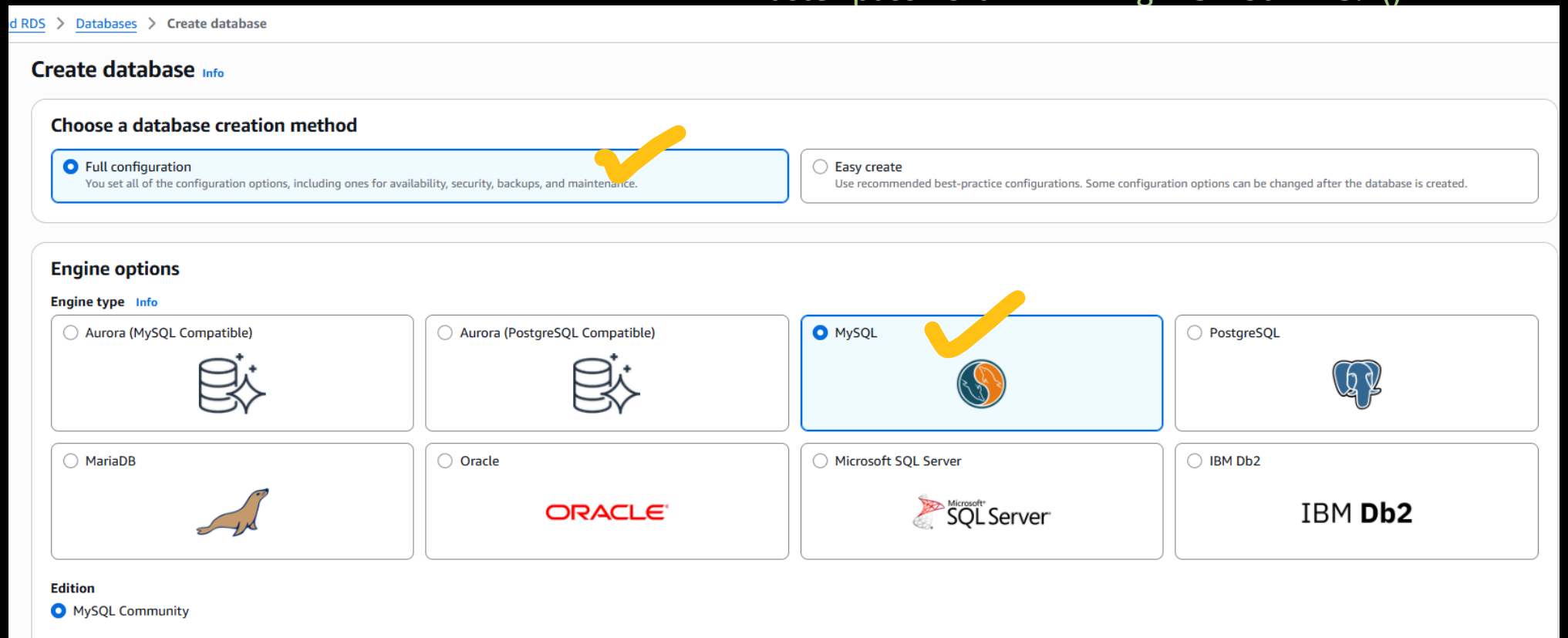
Learn how to create a database quickly with Easy create.

Start tutorial

Step 1: Create Database

Click **Create database.**
Choose following

Setting	Value
DB creation method	Full config(default)
Engine type	MySQL
Version	Latest
Template	Free tier
DB instance identifier	my-first-db
Master username	admin
Master password	gh45mod7=78%()



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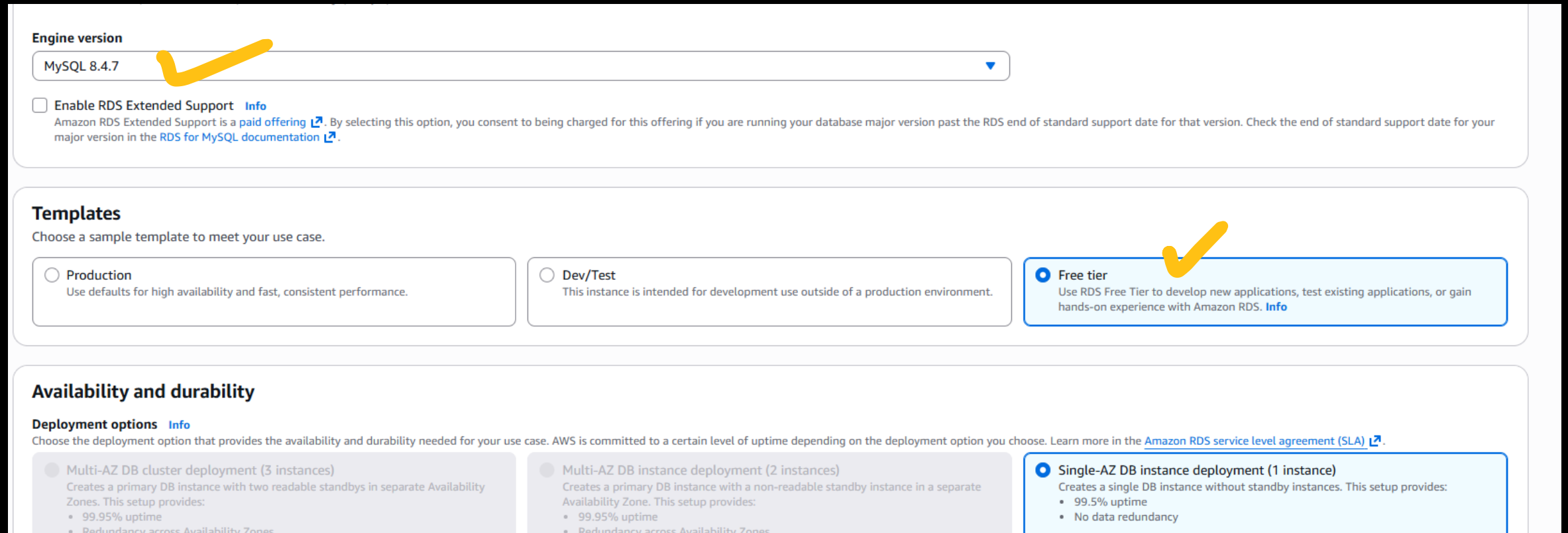
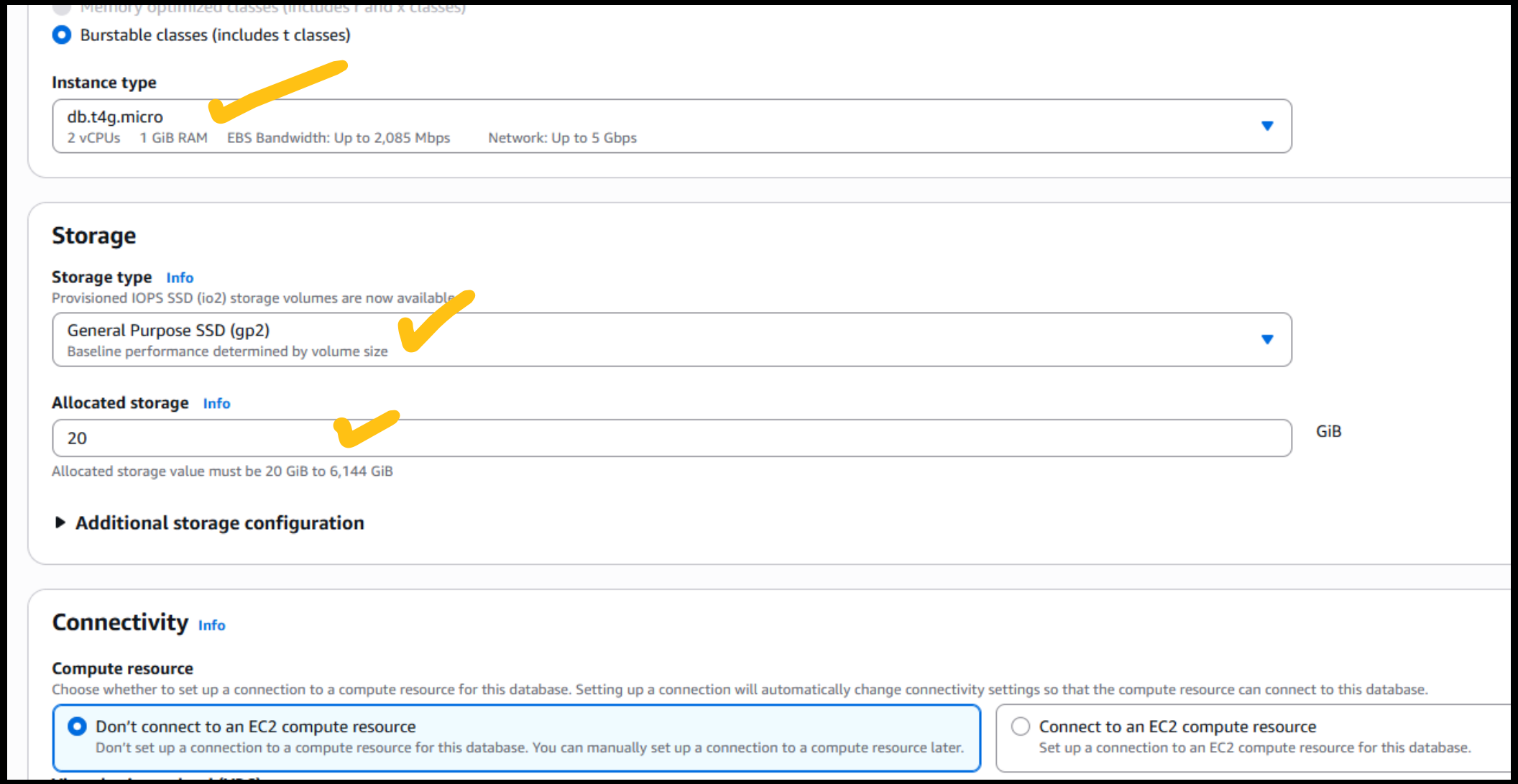


Step 2: Configure Instance

Recommended (Free tier):

Setting	Value
DB instance class	db.t3.micro
Storage	20 GB
Storage type	gp2



Step 3: Connectivity

Important settings:

Setting	Value
Public access	**Yes**
VPC security group	Default
Database port	**3306**

Public access Info

- (●) Yes
  RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or
  to the database.
- ( ) No
  RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resour

VPC security group (firewall) Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

- (●) Choose existing
  Choose existing VPC security groups
- ( ) Create new
  Create new VPC security group

Existing VPC security groups

Choose one or more options

default ×

Availability Zone Info

No preference

Step 3: Connectivity

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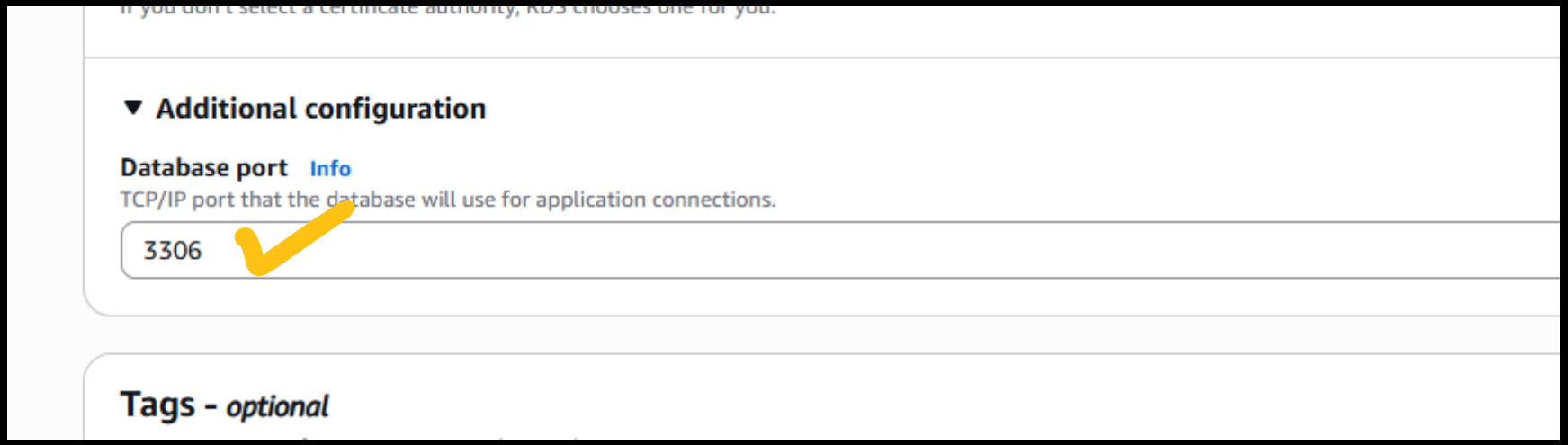

Step 4: Create Database

Click **Create database**
It will take about **3–5 minutes**.

2. Allow Your IP to Connect

Go to **RDS → Databases → my-first-db**

Open **Connectivity & Security -> Click Security Group**

Add inbound rule:

Type	Port	Source
MySQL/Aurora	3306	Your IP: Or use **0.0.0.0/0**

EC2 > Security Groups > sg-0234d66ab6d1d4c03 - default > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info		Description - optional Info	
sgr-04fe5ddfbdabebe0d	HTTP	TCP	80	Custom	0.0.0.0/0		Delete
sgr-01a77bd1fc46e5747	All traffic	All	All	Custom	sg-0234d66ab6d1d4c03		Delete
sgr-0f7e6cb124d040fa4	MYSQL/Aurora	TCP	3306	Custom	0.0.0.0/0		Delete

Add rule

Cancel | Preview changes | Save rules

3. Get code to connect to DB

Click on DB and from
Connect Using -> Code snippet,
get python code and run it.

You will use this in Python code to create DB first.

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrati

Connect using Info

- (●) Code snippets
  Use when connecting through SDK, APIs, or third-party tools including agents.
- ( ) CloudShell
  Use for a quick access to AWS CLI that launches directly fr Management Console.

Internet access gateway
Disabled

IAM Authentica
Disabled

Programming language
Python

Endpoint type
Instance endpoint

Connection steps
Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication confi

1. Install required packages

```
pip install mysql-connector-python boto3
```

2. Connection with DB Password

```
import mysql.connector
import boto3

password = "<Enter_DB_Password>"

conn = None
try:
```

4) On local Install Python MySQL Library

Use:

pip install mysql-connector-python boto3

pip install pymysql

5) The python code: Create table

```
import pymysql

connection = pymysql.connect(
    host="my-first-db.xxxxxx.us-east-1.rds.amazonaws.com",
    user="admin",
    password="gh45mod7=78%()",
    database=“student_db",
    port=3306
)

cursor = connection.cursor()
```

Cont…

```
create_table_query = """
CREATE TABLE IF NOT EXISTS students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100),
    course VARCHAR(100)
)
"""

cursor.execute(create_table_query)

connection.commit()

print("Table created successfully")

cursor.close()
connection.close()
```

6) The python code: Insert Data

```
import pymysql

connection = pymysql.connect(
    host="my-first-db.xxxxxx.us-east-1.rds.amazonaws.com",
    user="admin",
    password="gh45mod7=78%()",
    database="company_db"
)
cursor = connection.cursor()

query = """
INSERT INTO students (name, course)
VALUES (%s,%s)
"""

data = ("Alice", "Machine Learning")
cursor.execute(query, data)
connection.commit()
print("Data inserted")

cursor.close()
connection.close()
```

Cont…

6) The python code: **Fetch** Data

```
import pymysql

connection = pymysql.connect(
    host="my-first-db.xxxxxx.us-east-1.rds.amazonaws.com",
    user="admin",
    password="gh45mod7=78%()",
    database="company_db"
)
cursor = connection.cursor()

cursor.execute("SELECT * FROM students")

rows = cursor.fetchall()

for r in rows:
    print(r)

cursor.close()
connection.close()
```

Cont…

Most ML pipelines use this pattern:

Python ETL

↓

AWS RDS (MySQL/Postgres)

↓

Data warehouse / analytics

Cont…

Tect box